Hussein Alali Alismael

Backend Developer (Python/Django) | ML/DL Engineer

+963 939 866 304 | husseinalalialismael@gmail.com | https://www.linkedin.com/in/hussein-alali-alismael10 | https://github.com/hussein-alali-alismael | Aleppo, Syria

Professional Summary

Highly motivated 4th-year Informatics Engineering student specializing in **Artificial Intelligence**, with a strong focus on **Backend Development** and **Scalable APIs**. Proficient in building robust server-side logic using **Django REST Framework** and integrating **Neural Networks** into functional systems. Experienced in hardware-software integration using **Raspberry Pi** and passionate about solving complex engineering problems through optimized code and logical design.

Technical Skills

- **Backend Development:** Python, Django, Django REST Framework (DRF), RESTful APIs, Secure Authentication.
- **Artificial Intelligence:** Machine Learning (ML), Deep Learning, Neural Networks, Computer Vision, TensorFlow, PyTorch, XGBoost.
- **Database Management:** SQL, PostgreSQL, Database Design, Query Optimization.
- **Embedded Systems:** Raspberry Pi 4, Hardware Integration, Sensor Interfacing, GPIO Programming.
- **Tools & Environment:** Git, GitHub, Linux (Ubuntu), VS Code, Jupyter Notebooks, Logic Programming (Prolog).

Professional Projects

Rainfall Anomaly Explorer: ML for Humanitarian Support

- **Engineered** an end-to-end machine learning system to predict rainfall anomalies in Syria, achieving a **99.3% accuracy** ($R^2 = 0.9934$) using **XGBoost**.
- **Processed** and analyzed over 11,000 satellite-derived observations to provide actionable insights for humanitarian aid and disaster preparedness.
- **Implemented** SHAP-based interpretability to explain model predictions, ensuring transparency for humanitarian stakeholders and decision-makers.
- **Technologies:** Python, XGBoost, Scikit-learn, SHAP, Pandas, Matplotlib.

Team Management System API (Backend)

- **Developed** a secure backend infrastructure for team management using **Django REST Framework**.
- **Designed** and implemented RESTful APIs for task assignments, user authentication, and hierarchical team structures.
- **Managed** database schemas and **optimized** PostgreSQL queries to enhance data retrieval performance and system scalability.

Agricultural AI Robot (Olive Fruit Fly Detection)

- **Engineered** and trained a Neural Network model (GoogLeNet) to detect "Olive Fruit Fly" infestations from image datasets.
- **Implemented** advanced image processing techniques to improve model accuracy and robustness across various environmental conditions.
- **Integrated** the AI model with backend services to provide real-time detection capabilities.

Robotics & Hardware Integration (Raspberry Pi 4)

- **Developed** custom robotic prototypes by bridging software logic with physical hardware control using **Raspberry Pi 4**.
- **Programmed** sensor data collection and motor control sequences, ensuring seamless communication between sensors and the central controller.

Education

Bachelor of Informatics Engineering – Specialization in AI *Aleppo University, Syria* | [2022 – Present]

- **Relevant Coursework:** Machine Learning, Deep Learning, Database Systems, Algorithm Analysis, Robotics, Data Structures.

Certifications & Courses

- **Artificial Neural Networks** – projects.sy (40 hours)
- **Python Programming Language** – Midad Organization (35 hours)
- **Machine Learning** – Midad Organization (30 hours)

Soft Skills

- **Analytical Thinking:** Expert at breaking down complex requirements into manageable technical tasks.

- **Problem Solving:** Strong ability to debug complex logic and optimize system-wide performance.
- **Continuous Learning:** Proactively staying updated with the latest trends in AI and backend architectures.